Data Analysis Sheet – Reverse Genetics Experiment for AN8455

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Gene Information

- **Gene Name:** Uncharacterised
- **Systematic Name:** AN8455, ANID_08455
- **Gene Type:** Protein Coding Gene
- **Description:** AN8455 (ANID_08455) is a protein-coding gene in Aspergillus nidulans FGSC A4 strain located on the negative strand of chromosome V (541006–543575). Orthologs are present in several Aspergillus species, indicating conservation within this lineage. No specific molecular function has yet been assigned in the annotation. However, transcriptomic data suggest condition-dependent regulation, including differential expression across nitrogen sources, developmental stages, and selected regulatory mutant backgrounds. The gene contains 8 exons and 1 annotated transcript (AN8455-T), encoding a protein of 653 amino acids, which is predicted to be intracellular, non-secreted, and non-transmembrane and most likely localized to the cytoplasm or nucleus. Overall, AN8455 appears to be a conserved but uncharacterised fungal gene whose role requires experimental validation by reverse genetics.
- **Reference:** For A. nidulans, no gene-specific literature is available.
- **Genome Location:** ChrV_A_nidulans_FGSC_A4, 541006 - 543575 in Reverse Strand
- **Annotated DNA Sequence:** Annotated sequence can be retrieved in Appendix II, where 1200 bp upstream and downstream the gene were included.
- **Orthologs:**
  - A. fumigatus Af293: Afu5g10940, Afu5g10960
  - Neosartorya fischeri NRRL 181: NFIA_076300
  - Aspergillus wentii: Aspwe1_0041244
  - Aspergillus versicolor: Aspve1_0042580

Transcriptomics Information

- **Transcript:** 1 transcript (AN8455-T): an mRNA formed by 8 exons with a total length of 1980 nucleotides.
- **Transcriptomics (RNA-seq) Analysis:**
  - AN8455 is differentially expressed across conditions:
  - Ammonium-rich conditions: significantly upregulated – nitrogen-linked regulatory.
  - Deletion of veA or mtfA genes: upregulated – pathways linked to regulatory.
  - Complete media conditions: significantly upregulated – sensitivity to nutrition supply.
  - Development phase: upregulated in hyphae relative to conidia – related to fungi growth.
  - Crystalline cellulose exposure: moderate downregulation – independent of carbon metabolism.
- **Functional Analysis (GO Annotation Sourced from AspGD):**
  - GO:0008150: biological_process
  - GO:0005575: celluar_component
  - GO:0003674: molecular_function
  - Only three root terms of GO annotation are listed.

Protein Information

- **Predicted Protein Sequence:**

MSDQYSLSHLLEQQHLADCTKYERQWRPSDDNSPPPNEPVGPGTMRDVREEQFYLGGKNTVQPWDDSFP
SPLMPAIYDPYPDYNSGVWSDTWKGTFRTCEGPRGGTLDRQNVEDMVLTYPGVQGGFPYPKYGSYEALGL
DGYTCTTRASRLAAYGYSESGLNSTADPIVWDNVNWGTLQSECLQRNINRYHEPKRTPRLSILPLNPIRTEAPA
NDDEGISSPATKKRTAVVLRAWHDIVWTENLKEHVRSLIMELSLHSGAEYEVFLLTHVKDNDLLLCGVDEAKI
RALKKKFIPREFWDITVLFDERTLQSWYPKVDEHRPVYQYAQPLQIFSTVFPDFHYYWQLEMDARFTGHTY
HFLERASEFAKQQSRKYLWERNAYFYIPGAHGTWEEFMHTVDKSLLKRESNTVWGPAPPHSWIQPVGPEPP
ISSPQDDDYEWGVGEEADLITFLPIFDPSDTTWVFWNLLWNLPIETTTRRTSPVMIGRFSHRLLQVMHNAQI
EQGVALVSEMMPSSFALWHGLKAVHAPHPIYADGKWMPKELNNIVNKVAGGPEKINGGKDSFWNWDHKL
DHIVYRMSYMFTGQPAEDFYRRWLGYKPNPAQYTDGSRDPQGRNWFDGGHLREDIYGPLCFPSMLLHPIK
NTELDKGPDMAVPV

- **Protein Features and Properties:**
  - Length: 653 Amino Acids
  - Isoelectric Point: 5.75
  - Molecular Weight: 75391 Da
  - no Signal Peptide (SignalP)
  - no transmembrane helix (TMHMM)
- **Detailed Test Results:**
  - **Pfam Domain:** One major conserved domain (PF11885 – DUF3405 Protein of unknown function DUF3405) located at amino acids 143–640 with highly significant matches (E-value = 0) and corresponds to InterPro entry IPR021822.
  - **GPI-anchor prediction (big-PI Predictor in Fungi):** no potential GPI-modification site – no glycosylphosphatidylinositol (GPI) anchored surface or cell-wall-associated role for AN8455.
  - **WoLF PSORT:** most likely cytoplasmic, although a possible nuclear localisation was also suggested (queryProtein WoLFPSORT prediction cyto: 10.5, nucl: 9, cyto_pero: 8, pero: 4.5, mito: 2, vacu: 1).
  - **MitoProt:** no predictable mitochondrial targeting sequence (PROBABILITY of export to mitochondria: 0.0646) –unlikely to be imported into mitochondria (consistent with WoLF PSORT score for mito).
  - The protein encoded by AN8455 is predicted to be intracellular, non-secreted, and non-transmembrane.
- **Predicted Structure (AlphaFold 3):**
  - Predicted structure can be seen in Figure 7A.
  - Overall pTM score is 0.93 – high confidence for the structural accuracy

Af-pyrG Information

- **Annotated DNA Sequence:** Annotated sequence can be retrieved in Appendix II
- **Predicted Protein Sequence:** 278 Amino Acids -
MSSKSQLTYGARASKHPNPLAKRLFEIAEAKKTNVTVSADVTTTRELLDLADRLGPYIAVIKTHIDILTDFSVD
TINGLNVLAQKHNFLIFEDRKFIDIGNTVQKQYHGGALRISEWAHIINCSVLPGEGIVEALAQTASAQDFPYG
PERGLLVLAEMTSKGSLATGEYTKASVDYARKYKNFVMGFVSTRALTEVQSDVSSASEDEDFVVFTTGVNLSS
KGDKLGQQYQTPASAIGRGADFIIAGRGIYAAPDPVEAAQRYQKEGWEAYMARVCGKS
- **Predicted Structure (AlphaFold 3):**
  - Predicted structure can be seen in Figure 7B.
  - Overall pTM score is 0.95 – high confidence for the structural accuracy
- **Expected PCR Product Size:** 1124 bp

Experiment Related Information

- **Primers for Deletion Cassette:**
  - Blue sequences are consensus sites.
  - Black sequences are AN8455-specific deletion cassette primers.
  - 5’ Forward: *5’-GTAACGCCAGGGTTTTCCCAGTCACGACGGGGTGAAGGGATACAGTTAC-3’*
  - 5’ Reverse: *5’-ATCCACTTAACGTTACTGAAATCGATCAGACATGGAGGACAC-3’*
  - 3’ Forward: *5’-CTCCTTCAATATCATCTTCTGTCCTGATATGGCAGTCCCAGT-3’*
  - 3’ Reverse: *5’-GCGGATAACAATTTCACACAGGAAACAGCAGGAGTGTAAGGCTGAAGAG-3’*
- **Primers for Diagnosis:**
  - ANID_08455 Forward: *5’-TAAGGCGAGAATAAAGAAA-3’*
  - ANID_08455 Reverse: *5’-TTAGTGATAGAGGCGAAGAG-3’*
- **Expected PCR Product Size:** 1124 bp

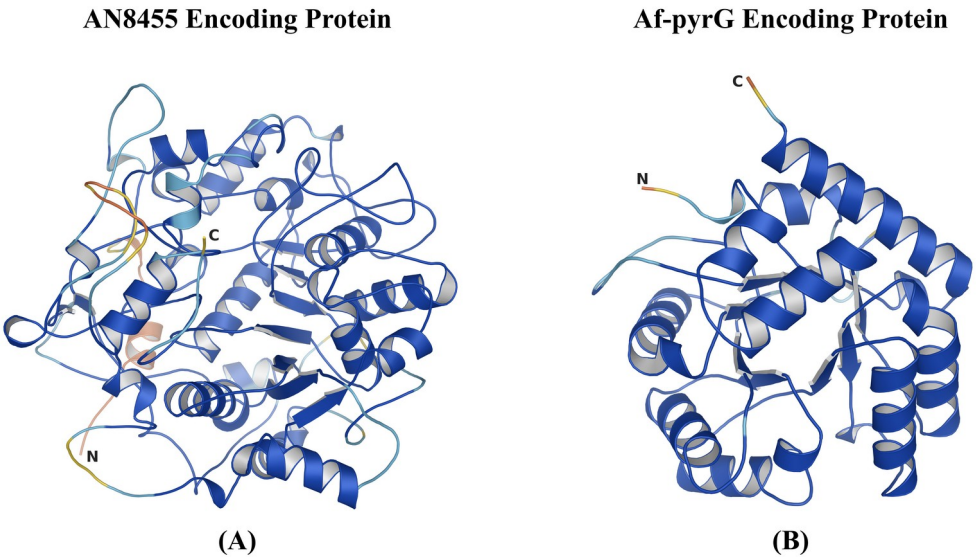


Figure 7. *AlphaFold3 predicted protein structures. Colored by confidence level of each residue: dark blue for very high confidence, light blue for high, yellow for low, and orange for very low.* ***(A)*** *Protein encoded by AN8455 (Overall pTM score: 0.93).* ***(B)*** *Protein encoded by Af-pyrG+ (Overall pTM score: 0.95).*